



Global Superstore Sales & Profitability Analysis

An End-to-End Business Intelligence Project

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Executive Summary & Methodology

Objective

End-to-end EDA to uncover insights on sales, profitability, customers, products, regions, shipping, promotions, and returns.

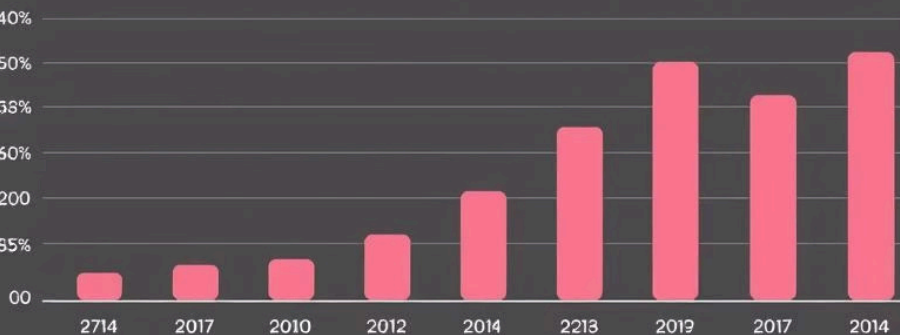
Tools Utilised

Python, Pandas, Seaborn, Statsmodels, Clustering for robust analysis.

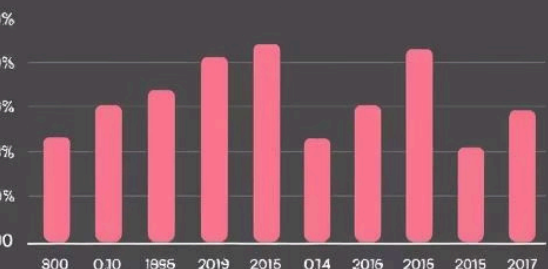
Key Deliverables

Notebooks, Markdown Reports, Visualisations, and actionable Recommendations.

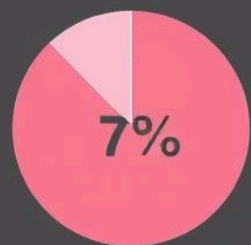
New percent of coliforms recontaminated



Quarterly loss contribution



Order: both orders



Key Business Insights



Sales Growth Trends

Strong Q4 seasonality drives annual sales peaks.



Discount Impact

Discounts exceeding 20% severely erode profitability.



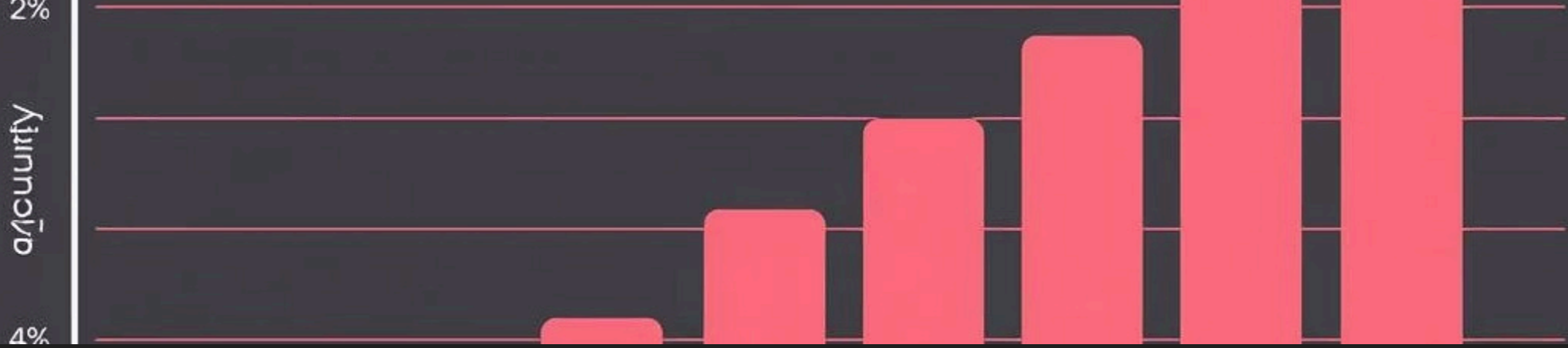
Loss-Making Orders

Approximately 25% of all orders generate losses.



Customer Value

Loyal customers contribute the majority of revenue.



Profitability & Promotion Insights

Overall Margin

Average profit margin stands at 11.6%.

Discount Threshold

Discounts over 30% consistently result in negative margins.

Discount Elasticity

Every 10% discount increase reduces profit by approximately £24.

Loss Leaders

Top 20 SKUs account for 80% of all negative profit.

Customer & Product Analysis



RFM Segmentation

Customers categorised into Loyal, Champion, At-Risk, and Lost segments.



KMeans Clustering

Identified 4 distinct customer behavioural segments.



Product Lifecycle

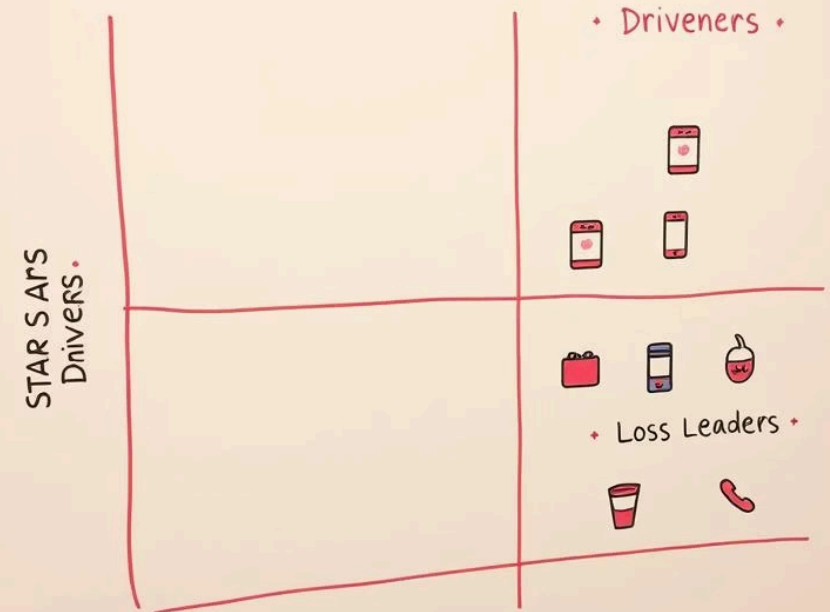
Products classified as Stars, Drivers, or Loss Leaders.



Strategic Focus

Prioritise high-margin, high-retention SKUs for growth.

Product Matrix





Regional & Shipping Dynamics

Top Markets

Western Europe, West US, and India are the highest performing regions.

Shipping Costs

Standard Class shipping is dominant but incurs high costs, negatively correlating with profit.

1

2

3

Risk Zones

Oceania and Western Asia exhibit lower profitability.



Discount threstold

7 Logistics

Strategic Recommendations



Cap Discounts

Implement a strict 20% maximum discount policy.



Address Loss Leaders

Eliminate or reprice unprofitable products.



Invest Strategically

Focus resources on profitable regions and loyal customer segments.



Optimise Logistics

Enhance shipping strategies to reduce costs.



Closing & Next Steps

Build ML Models

Predict profit and return probability using advanced analytics.

Deploy Dashboards

Implement real-time Business Intelligence for continuous monitoring.

Extend Optimisation

Expand to forecasting and product-level optimisation strategies.

Thank you for your attention. Questions?